

Exercise 1 — Regex: Multi-Rule String Pattern

Rules

- Starts with 1–3 occurrences of the character 'S'.
- Followed by at least 3 occurrences of the substring "com".
- Followed by 0 or 1 occurrence of the character '%'.
This is what the rule means by "the group (@#\$)" — the parentheses mark it as one unit to be repeated, not a set of interchangeable characters.
- Ends with 2 occurrences of the literal 3-character sequence "@#\$" (i.e. the string ends with "@#\$@#\$"). This is what the rule means by "the group (@#\$)" — the parentheses mark it as one unit to be repeated, not a set of interchangeable characters.

Examples

- Valid: `Scomcomcom@#$@#$` (one S, three coms, no %, ends correctly).
- Valid: `SSScomcomcomcom%@#$@#$` (three S's, four coms, with %, ends correctly).
- Invalid: `SScomcom@#$@#$` (only two coms — rule 2 needs at least three).
- Invalid: `Scomcomcom@#$` (only one @#\$ at the end — rule 4 needs exactly two).
- Invalid: `SSSScomcomcom@#$@#$` (four S's — rule 1 allows at most three).

Exercise 2 — Regex: Vietnamese Phone Number

Rules

- Starts with either "0" or "+84".
- Followed by exactly 9 digits (regardless of which prefix was used).

Examples

- Valid: `0912345678` ("0" + 9 digits).
- Valid: `+84912345678` ("84" + 9 digits).
- Invalid: `84912345678` (missing the '+' before 84, so it doesn't match either accepted prefix).
- Invalid: `091234567` ("0" + only 8 digits).
- Invalid: `+841234567890` ("84" + 10 digits, one too many).

Exercise 3 — Regex: Parenthesized Area Code + Number

Rules

- The string has the literal form `(local)number` — an opening parenthesis, the local code, a closing parenthesis, then the number, with nothing else around them.
- local: 2–4 digits total, and the first digit must be 0.
- number: 5–7 digits.

Examples

- Valid: `(024)1234567` (local "024", 3 digits starting with 0; number "1234567", 7 digits).
- Valid: `(08)12345` (local "08", 2 digits; number "12345", 5 digits).
- Invalid: `(24)1234567` (local doesn't start with 0).
- Invalid: `(02468)1234567` (local is 5 digits — rule allows at most 4).
- Invalid: `024)1234567` (missing the opening parenthesis).

Exercise 4 — Regex: Contains 'a' or 'b', but Never "ab"

Rules

- The string must contain at least one 'a' or at least one 'b' (having both, or just one of them, is fine).
- The string must never contain the substring "ab" (an 'a' immediately followed by a 'b'), anywhere.

Examples

- Valid: **ba** (contains 'a' and 'b', but 'b' comes before 'a' — never "ab").
- Valid: **aa**, **bb**, **cba** (contain 'a'/'b', no "ab" substring).
- Invalid: **ab** (is exactly the forbidden substring).
- Invalid: **cab**, **aabb** (both contain "ab" as a substring somewhere).
- Invalid: **xyz** (contains neither 'a' nor 'b', so it fails the first rule even though it also has no "ab").

Exercise 5 — Student ID Validation

Format

- The ID is **Bxxxxxxx** — a literal 'B' followed by exactly 7 digits, where each digit must be 1–9 (0 is not allowed in any of the 7 positions).

Program requirements

- Prompt the user to enter a student ID from the keyboard.
- Check it against the required format.
- Print a clear result — e.g. "Valid student ID" or "Invalid student ID".

Examples

- Valid: **B1234567** (7 digits, all 1–9).
 - Invalid: **B1234560** (contains a 0).
 - Invalid: **A1234567** (wrong leading letter).
 - Invalid: **B123456** (only 6 digits).
 - Invalid: **B12345678** (8 digits, one too many).
-